Himanshu Kumar

9525276093 | himanshukumar952527@gmail.com | https://www.linkedin.com/in/himanshu-kumar-8a312236a/ |
https://github.com/himanshukumar952527-a11y

Education

Jawaharlal Nehru University — New Delhi, Delhi
Bachelor of Technology in Electronics and Communication Engineering — *2025-2029*

H/S Khairwa Bishnupur — Sitamarhi, Bihar
Intermediate in Science — *2022-2024*

Experience

Self-Learning and Project Development — 2025 – Present
Artificial Intelligence and Programming — *Remote*

- Worked on personal projects related to data analysis and algorithm implementation using Python and C++
- Explored Machine Learning concepts and libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn
- Practiced Data Structures and Algorithms through coding platforms like LeetCode

Projects

Loan Approval Prediction System | *Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn* — 2026

- Developed an end-to-end machine learning system to predict loan approval decisions using demographic, financial, employment, and credit-related applicant data
- Performed comprehensive data preprocessing including missing value treatment, feature encoding, feature scaling, outlier detection, and exploratory data analysis (EDA)
- Implemented and compared multiple classification algorithms including Logistic Regression, KNN, Naive Bayes, Decision Tree, SVM, and Random Forest
- Applied 5-Fold Cross Validation and GridSearchCV for hyperparameter tuning, selecting Random Forest as the best-performing model with **93% Accuracy**, **0.8923 F1-Score**, and **0.9166 Cross-Validation F1-Score**
- Evaluated models using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and Train-Test performance analysis to ensure robust generalization

Employee Attrition Prediction | *Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn* — 2026

- Built a Logistic Regression-based classification model to predict employee attrition using HR workforce data
- Conducted extensive EDA, feature engineering, correlation analysis, and categorical encoding to prepare data for modeling
- Implemented K-Fold Cross Validation and hyperparameter tuning to enhance model reliability and reduce overfitting
- Assessed model performance using ROC-AUC Score, Precision-Recall Analysis, Accuracy, Precision, Recall, F1-Score, and Confusion Matrix
- Created ROC and Precision-Recall Curves to analyze classification thresholds and support data-driven HR retention strategies

Analysis of Rainfall Across India | *Python, Pandas, NumPy, Matplotlib,Seaborn* — 2026

- Analyzed rainfall datasets from different regions of India to identify seasonal and regional trends
- Performed data cleaning and visualization using Python libraries for comparative analysis
- Generated graphical insights to study rainfall distribution patterns across states
- Improved understanding of real-world climate data through statistical analysis techniques

Maze Generator and Solver | *C,Python, DFS, BFS, Data Structures* — 2026

- Developed a maze generation and solving system using graph traversal algorithms
- Implemented DFS and BFS algorithms for pathfinding and shortest path detection
- Designed logic for efficient maze creation and navigation visualization
- Strengthened problem-solving and algorithmic thinking through practical implementation

Technical Skills

Languages: C, Python, HTML, CSS
Core Concepts: Data Structures and Algorithms, Machine Learning
Developer Tools: VS Code, Jupyter Lab, Git
Libraries: Pandas, NumPy, Matplotlib, Scikit-learn
Areas of Interest: Machine Learning , Backend Development,Agentic AI,Generative AI, Computer Vision